

Programme: B.Sc.DS.

Division: _____

Roll No. _____

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

Date : 03/02/2025

Mid Term Exam February 2025

Time: 1hr

T.Y. B. Sc. IT/DS.

SEM -VI

Subject: Exploratory Data Analysis

Marks: 20

Instructions: All questions are compulsory.

Bloom's Taxonomy Level	CO	Q1	Attempt any Two	Marks
2	CO1	(a)	Compare structured, unstructured and semi-structured data.	5
5	CO1	(b)	Discuss the different aspects of data.	5
3	CO1	(c)	What is a statistic? Explain the various types of statistic.	5
6	CO1	(d)	Discuss a few applications of Business Statistics.	5
Bloom's Taxonomy Level	CO	Q2	Attempt any Two	Marks
5	CO2	(a)	With the help of an example, highlight the significant role of EDA.	5
2	CO2	(b)	Compare EDA with Bayesian data analysis, with the help of an example.	5
5	CO2	(c)	Explain with the help of an example, how lag plot helps in drawing insights from data.	5
6	CO2	(d)	Illustrate with an example how graphical representation overcomes the limitations of quantitative analysis.	5

Programme: B.Sc.DS.

Division: _____

Roll No. _____

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

Date : 03 / 02 / 2025

Mid Term Exam February 2025

Time: 1hr

T.Y. B. Sc. DS

SEM -VI Subject: Machine Learning

Marks: 20

Instructions: All questions are compulsory.

Bloom's Taxonomy Level	CO	Q1	Attempt any Two	Marks 10
2	1	(a)	Explain the various applications of Machine Learning.	5
3	1	(b)	Draw a general architecture of ML systems and explain each phase.	5
2	1	(c)	Explain the concept of Generalization in Machine Learning	5
3	1	(d)	Describe the workflow of Unsupervised Learning Model.	5
Bloom's Taxonomy Level	CO	Q2	Attempt any Two	Marks 10
2	2	(a)	Explain the different methods of computing distance in KNN model.	5
2	2	(b)	Write a short note on Multi-Class Classification in Machine Learning.	5
3	2	(c)	List and explain the steps involved in ID3 Algorithm.	5
3	2	(d)	Describe the concept of Attribute selection Method – Gini Impurity.	5

Programme: B.Sc.DS.

Division: _____

Roll No. _____

Date : 04/02/2025

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

Mid Term Exam February 2025

T.Y. B. Sc. DS.

SEM -VI

Subject: Applied Business Analytics

Time: 1hr

Marks: 20

Instructions: All questions are compulsory.

Bloom's Taxonomy Level	CO	Q1	Attempt any Two	Marks
1	1	(a)	Define business analytics. Explain its necessity in Data analytics.	10
3	1	(b)	What is the purpose of trendlines in Excel? Explain with an example.	5
1	1	(c)	What are the empirical rules for interpreting std. deviation?	5
2	1	(d)	What are the measures of shape? Explain.	5
Bloom's Taxonomy Level	CO	Q2	Attempt any Two	Marks
3	2	(a)	Explain in brief Business Intelligence tools used for DA.	10
5	2	(b)	Write a short note on conditional mean variance.	5
4	2	(c)	Differentiate between objective and subjective probabilities.	5
1	2	(d)	What are the elements of Decision Analysis? Explain.	5

Programme: B.Sc.D.S

Division: _____

Roll No. _____

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

Date: 04/02/2025

Mid Term Exam February 2025

Time: 1hr

T.Y.B.Sc.DS. SEM – VI

Subject: Internet of Things

Marks: 20

Instructions: All questions are compulsory.

Bloom's Taxonomy Level	CO	Q1	Attempt any Two.	Marks 10
6	1	(a)	"Any sufficiently advanced technology is indistinguishable from magic". Discuss.	5
2	1	(b)	What is manufactured normalcy field? Explain.	5
4	1	(c)	Differentiate between TCP and OSI layer	5
1	1	(d)	Define IoT. Explain ubiquitous computing.	5
Bloom's Taxonomy Level	CO	Q2	Attempt any Two.	Marks 10
2	2	(a)	How is development done for Arduino? Explain.	5
6	2	(b)	Discuss the factors we should be consider when deciding to build Internet of Things device.	5
6	2	(c)	"Open source has a competitive advantage". Discuss.	5
2	2	(d)	Explain Sensors and actuators.	5

Programme: B.Sc.DS.

Division: _____

Roll No. _____

Date: 05 / 02 / 2025

Y. B. Sc. D.S.

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

Mid Term Exam February 2025

SEM - VI

Subject: Healthcare Analytics

Time: 1hr

Marks: 20

Instructions: All questions are compulsory.

Bloom's Taxonomy Level	CO	Q1	Attempt any Two (From Unit 1)	Marks
				10
2	1	(a)	Summarize the Mathematics domain in Healthcare Analytics.	5
6	1	(b)	What are steps to download and use SQLite?	5
3	1	(c)	How is financing done in healthcare?	5
6	1	(d)	Explain about the history of healthcare analytics.	5
Bloom's Taxonomy Level	CO	Q2	Attempt any Two (From Unit 2)	Marks
				10
6	2	(a)	How will you calculate the baseline myocardial infarction (MI) probability?	5
6	2	(b)	How will you use Criterion tables and the weighted sum approach.	5
2	2	(c)	Explain the Model frameworks for medical decision making.	5
2	2	(d)	Describe the various machine learning algorithms used in Healthcare Analytics.	5